YESSINE ABDELMAKSOUD

AI Engineer | Data Engineering

Engineering intelligent systems from data to production-ready AI solutions.

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01 PROFILE

AI Engineer with a Data Engineering background focused on designing and building end-to-end AI systems. From data processing and machine learning to LLM agents, APIs, and deployment, I combine applied data science with solid software architecture to build reliable, scalable, and production-oriented solutions.

02 EXPERIENCE

Software Developer / AI Engineer Jun 2026–Present
YouTestMe · Remote Remote

- Modernizing the production **GetCertified** platform by rebuilding core test-taking workflows in **React**, integrating Java REST APIs, and preserving legacy compatibility.
- Developing a **centralized administration dashboard** that consolidates data across multiple product instances for unified monitoring and analytics.
- Developing a **RAG powered knowledge assistant** that retrieves information from internal documentation and run-books, providing contextual answers with **source citations**.
- Building **ML based usage and capacity forecasting** to predict storage consumption, testing activity, and concurrent users from historical data, supporting proactive resource planning.

AI / DevOps Intern Aug 2025 & Jun 2026
KPIT Technologies · Sfax, Tunisia On-site

- Architected an **AI-driven CI/CD remediation platform** for embedded software using three agents for **failure diagnosis, patch generation, and code review**.
- Engineered automated root-cause analysis and patch validation with **LangGraph, FastAPI, and GitLab CI**, including human approval before merge.
- Integrated **K3s, Redis, MinIO, Ollama/Claude**, and **Prometheus/Grafana/Loki** for orchestration, reliability, and observability.

AI / DevOps Intern Aug 2025
Primatec · Sfax, Tunisia On-site

- Developed a **LangGraph-based CI/CD assistant** to analyze **Jenkins build failures** and generate AI powered root cause diagnoses and troubleshooting recommendations.
- Implemented **real time pipeline monitoring**, build failure notifications, automated retries, and intelligent error-handling workflows.

Data Scientist Intern Jun–Jul 2025
Telcotec · Sfax, Tunisia On-site

- Built an **end-to-end ML solution** for indoor WiFi optimization, combining building-plan data extraction, predictive path-loss modeling, and access point placement optimization.
- Developed interactive **2D/3D coverage heatmaps** and multi-floor visualizations for wireless network planning.

03 SELECTED PROJECTS

AI-Augmented Security Operations Platform

Qwen3-14B · Ollama · Wazuh · Suricata · Elasticsearch · FastAPI

Built and deployed an **on-premises LLM-powered security analysis platform** in the faculty data center using **Qwen3-14B** to explain Wazuh and Suricata alerts, generate KQL investigations, and recommend **MITRE ATT&CK-aligned remediation**. Deployed across **5 VMs** and validated against **5 attack scenarios**.

LLM Fine-Tuning for 5G Network Diagnostics

Qwen2.5 · LoRA · Quantization · NLP · Python

Fine-tuned **Qwen2.5-1.5B-Instruct** using **LoRA and 4-bit quantization** to diagnose 5G throughput degradation from drive-test and gNodeB data. Achieved **53.82% accuracy** across 8 root-cause classes (12.5% random baseline) with only **1.18% trainable parameters**.

StressOFF · Multimodal AI Wellness Platform

Multimodal AI · LLM · FastAPI · Docker · Azure · Firebase

Built an **end-to-end multimodal AI platform** combining wearable data, meal images, and calendar context for stress and nutrition analysis with **LLM-powered conversational assistance**. Architected **4 FastAPI microservices** with streaming, output validation, and Firebase authentication; containerized with Docker and deployed to **Azure Container Instances**.

Electricity & Gas Fraud Detection Platform

Machine Learning · Python · Feature Engineering · APIs

Developed an **end-to-end fraud detection platform**, transforming raw utility billing data into predictive features, comparing classification models, and optimizing decision thresholds. Delivered **prediction APIs and interactive dashboards** with monitoring, alerting, and explainable fraud investigations.

04 TECHNICAL TOOLKIT

Languages	Python, SQL, C++,
AI / LLMs	RAG, LangGraph, AI Agents, Hugging Face, LoRA, Quantization, Embeddings, Reranking
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, Model Evaluation
Data Engineering	Pandas, NumPy, PostgreSQL, Spark, ETL, Data Warehousing, Power BI
Backend / MLOps	FastAPI, Docker, Kubernetes, GitLab CI, Jenkins, MLflow
Cloud / DevOps	Azure, Oracle Cloud, Linux, Git, Prometheus, Grafana, Loki

05 EDUCATION

Engineering Degree in Data Engineering

2024–2027 (Expected)
Final Year

Faculty of Sciences of Sfax · University of Sfax, Tunisia

Integrated Preparatory Cycle: 2022–2024 · Ranked **5th**.

06 CERTIFICATIONS, AWARDS & LEADERSHIP

CERTIFICATIONS

- **Microsoft:** Azure AI Fundamentals, Azure Data Fundamentals.
- **DeepLearning.AI:** Machine Learning Specialization, Artificial Intelligence Engineer 2.

HONORS & AWARDS

- **1st Place** – fit-tech, IEEE ENIS (2025).
- **1st Place** – Master Faster 2025, Google Developer Student Club FSS.
- **5th Place** – TSYB 13 AESS Challenge, IEEE (2025). Our solution : **CubeSat – Real-Time Anomaly Detection:** Developed an **Isolation Forest-based anomaly detection system** for CubeSat telemetry, integrating **Raspberry Pi/ESP32** data acquisition and Python monitoring to detect electrical anomalies (SEL events).

LEADERSHIP & COMMUNITY

Technical Manager – Google Developer Student Club, FSS Sfax *2025–2026*
Led a technical workshop on **MCP, ADK, and A2A**; organized hackathons and mentored students.